Dhirubhai Ambani University
Technology

Formerly DA-IICT

Exploratory Data Analysis of Daily Darshans at Tirumala Tirupati Devasthanam

A Partial Project [EDA] Report
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of the course [IT402 Applied_Forecasting_Methods]

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1. Abstract

The Tirumala Venkateswara Temple, which is managed by the Tirumala Tirupati Devasthanams, is a popular place that a lot of people visit. The Tirumala Venkateswara Temple attracts millions of pilgrims every year. This study looks at the number of people who visit the Tirumala Venkateswara Temple every day. The study uses data from October 2013 to July 2025. The data includes things like the weather, how many people visited the Tirumala Venkateswara Temple in the past, what people are searching for on Google, and whether it is a festival or holiday.

By looking at the data in ways this study finds some interesting patterns. For example, more people visit the Tirumala Venkateswara Temple on weekends and during the summer. The study also finds that the number of people visiting the Tirumala Venkateswara Temple can change from week to week. The Tirumala Venkateswara Temple has a lot of visitors during festivals like Brahmotsavam. The study also looks at what happened during the COVID-19 pandemic.

The things we learned from this study can help us understand how people behave when they visit the Tirumala Venkateswara Temple. This can help us make predictions about how many people will visit the Tirumala Venkateswara Temple in the future. These predictions can help the people in charge of the Tirumala Venkateswara Temple make decisions about managing the crowds, planning transportation and making money. The Tirumala Venkateswara Temple can use this information to make the experience better for the pilgrims who visit the Tirumala Venkateswara Temple.

2. Introduction

The Tirumala Venkateswara Temple, managed by the Tirumala Tirupati Devasthanams (TTD), is an important place for Hindus to visit. A lot of people, 60,000 to 80,000, go to the temple every day. More people visit during festivals, weekends, and holidays. So it's crucial to guess how many people will visit the temple each day. This helps in managing the crowd using resources properly and making the experience better for the visitors.

This report looks at the data on how many people visit the temple every day. We looked at datasets from different tourist places around the world. These places include.

Vaishno Devi Temple, Kedarnath, Goa, Bali, Jiuzhaigou Valley, South Korea, Las Vegas, Sri Lanka, Malaysia, and Thailand.

We chose the Tirumala Tirupati Devasthanams dataset because it has data and many details like weather, average visitors over time, Google Trends scores, and whether there are festivals or holidays. The other datasets we looked at mostly have data for months or years. Don't have enough details. This makes our chosen dataset perfect for studying and making predictions about how many people will visit the temple.

3. Dataset Description

3.1 Data Source

The primary dataset used in this study is the **Tirumala Tirupati Devasthanam Darshans Dataset**, publicly available on Kaggle. It contains daily records of pilgrim footfall at the Tirupati Balaji Temple from **16 October 2013 to 06 July 2025**.

Link:
https://www.kaggle.com/datasets/vishnumadhav2454/tirumala-tirupati-devasthanam-darshans-dataset

This dataset was chosen after reviewing more than 12 tourism and pilgrimage datasets from various locations across India and the world (detailed list and sources are attached in the Appendix). The Tirupati dataset stands out due to its daily granularity and comprehensive feature set.

3.2 Feature Description

The dataset consists of 16 features, including the target variable and various explanatory variables:

Target Variable:

- **darshans**: Daily number of pilgrims who received darshan at the temple (main variable of interest).

Key Features:

- **weekend**: Binary flag (1 = weekend, 0 = weekday)
- **summer**: Binary flag indicating summer vacation period
- **temp_max** & **temp_min**: Daily maximum and minimum temperature (°C)
- **humidity**: Daily average humidity (%)
- **rainfall**: Daily rainfall in mm
- **rolling_avg_8**: 8-day rolling average of darshans (captures recent momentum)
- **google_trend_score**: Google search interest score (0–100)
- **is_festival**: Binary flag for festival days
- **is_public_holiday**: Binary flag for public holidays
- **is_brahmostavam**: Binary flag for the annual Brahmostavam festival

4. Objectives of the Study

The main objectives of this Exploratory Data Analysis are:

1. The main goals of this Exploratory Data Analysis of the Tirumala Venkateswara Temple are:
2. To look at the distribution and patterns in daily visitor counts at the Tirumala Venkateswara Temple.
3. To find out what things affect how many people visit the Tirumala Venkateswara Temple, like the weather, weekends, festivals, and short-term trends.
4. To see if there are patterns in the data over time, like seasonality and long-term trends, and to find any anomalies, like what happened during the COVID-19 period.
5. To see how the different variables are related to each other using statistics and pictures, such as correlation analysis and scatter plots.
6. To get information that can help make good forecasting models to manage the crowds and plan operations at the Tirumala Venkateswara Temple.
7. To see which variables are most important for the daily visitor counts at the Tirumala Venkateswara Temple and to study how things that happened in the past affect the visitor patterns, at the Tirumala Venkateswara Temple, so we can make models of the data over time.

5. Data Preprocessing

The raw dataset went through steps to prepare it for analysis.

These steps included:

- Converting the date column to a date and time format.
- Setting the date as the reference point for time series analysis.
- Filling in missing values using a method that connects the dots to keep the data smooth.
- Creating some flags and averages where needed.

After looking at how all the features relate to each other

The following three features were removed to make the model less complicated and reduce repetition:

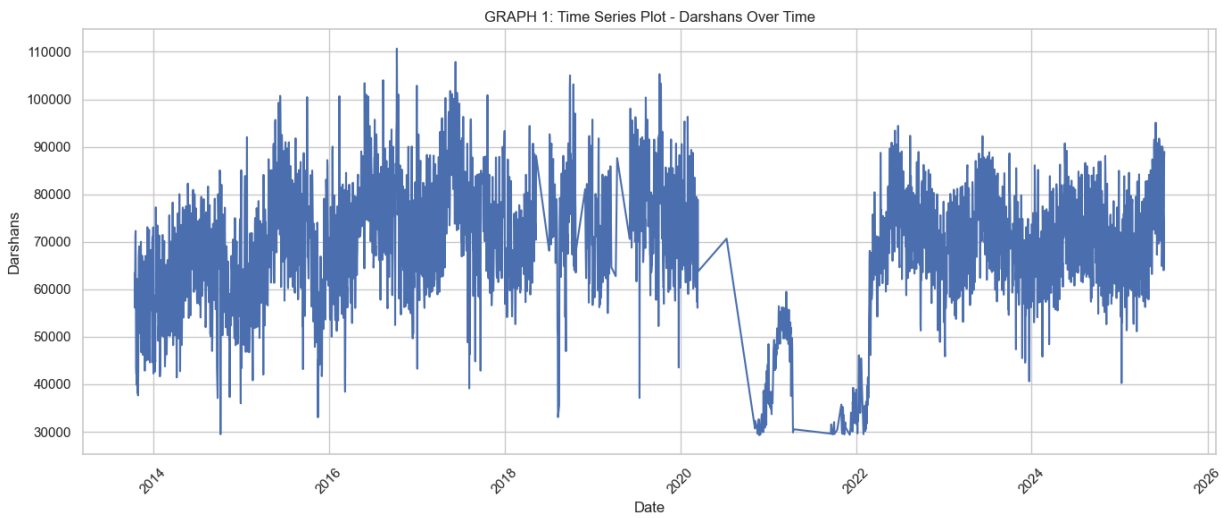
- Rolling_avg_7: This was removed because rolling_avg_8 is slightly better at predicting Darshans with a correlation of 0.73 compared to 0.72. Both rolling averages give information just one day apart, so keeping rolling_avg_8 avoids repetition while still capturing recent visitor trends.
- Weekday: This was removed because it is negatively correlated with weekend at −0.55. The weekend flag already captures the information, but in a positive way, with a correlation of 0.40 to Darshans. Including both would add repetitive information, so only the weekend was kept.
- is_public_holiday: This feature does not correlate well with darshans at 0.05. Its impact on visitor numbers is small compared to weekends, the summer season, and major festivals, making it less useful for prediction.

The broader festival flag and seasonal component already covers its influence.

Removing it helps simplify the model without losing much of its power.

6 . EXPLORATORY DATA ANALYSIS

6.1 Time Series Visualization



Manual Analysis:

1. Long-Term Trend Pattern

From 2014 to early 2020, Darshan counts show a generally increasing trend, indicating gradual growth in pilgrim arrivals over the years.

This suggests:

- Increasing tourism demand,
- Growing temple popularity,
- Improved accessibility or infrastructure.

2. Seasonal Repeating Fluctuations

The graph contains repeated up-and-down wave-like patterns throughout most years.

This indicates: ***Strong seasonality exists.***

Likely causes:

- Weekend crowd surges,
- Festival peaks,
- Holiday pilgrim inflow cycles.

The recurring spikes suggest regular behavioral tourism patterns.

3. Sudden Sharp Drop During 2020–2021

A dramatic collapse occurs around 2020–2021, where darshan counts fall close to 30,000.

This is a major structural break.

Most likely reason:

COVID-19 pandemic disruption

Possible causes:

- Temple closure restrictions,
- Lockdowns,
- Travel bans.

This period behaves abnormally compared to normal years.

4. Recovery Phase After 2021

From 2022 onward, the darshan counts rise again and stabilize around higher levels.

This indicates:

- Tourism recovery after the pandemic,
- Return of pilgrim traffic,
- Restoration of normal seasonal behavior.

5. Presence of Extreme Peaks and Dips

Several sudden spikes exceed 100,000 darshans.

These may correspond to:

- Major temple festivals,
- Brahmotsavam periods,
- Public holiday rush days.

Similarly, sudden dips may represent:

- Weather disruptions,
- Restricted access days,
- Exceptionally low-demand events.

Key Insights:

Important Findings:

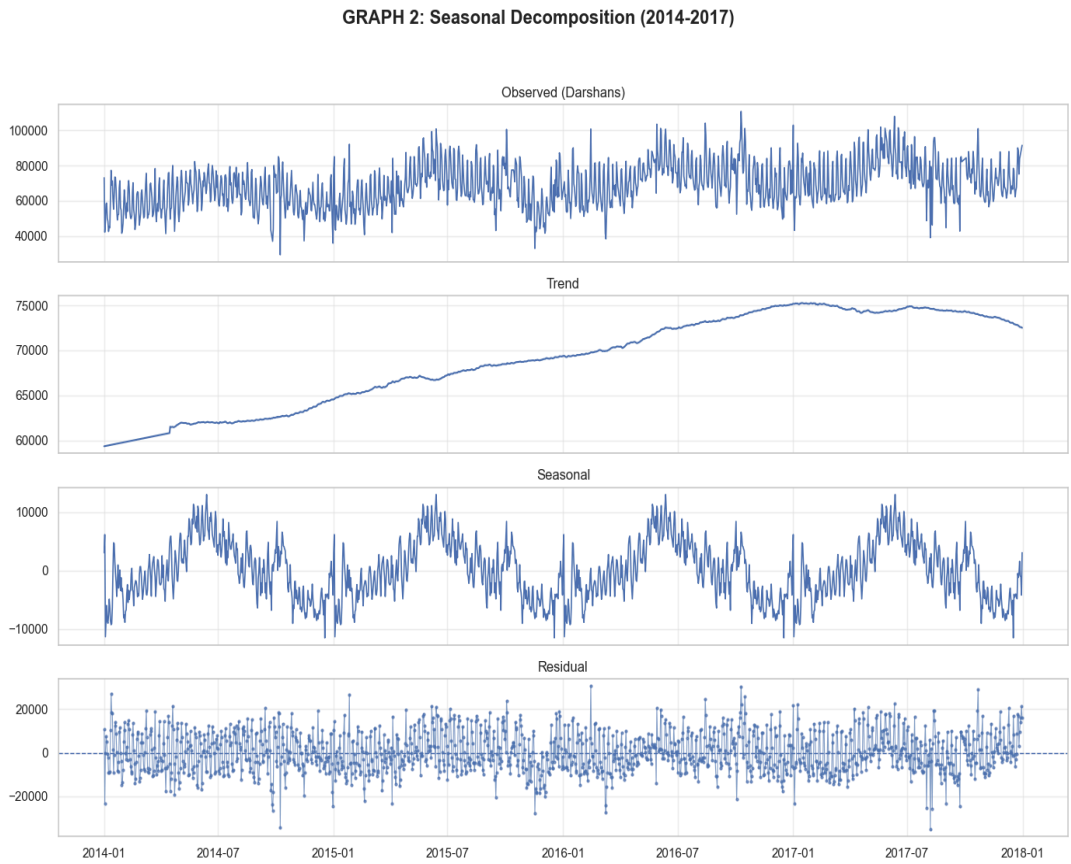
- Strong long-term tourism growth trend before COVID.
- Clear seasonal recurring patterns.
- COVID caused severe disruption.
- Strong recovery visible after the pandemic.
- Demand remains highly variable over time.

Conclusion:

The time series plot reveals that pilgrim arrivals are highly seasonal and trend-driven, with a major external disruption during the COVID period. The series demonstrates strong forecasting potential because clear temporal patterns are visible.

6.2 Seasonal Decomposition Analysis

6.2.1 Seasonal Decomposition Analysis (2014–2017)



The **Observed** series exhibits high daily volatility, with visitor counts ranging between approximately 40,000 and 105,000. A gradual upward movement is visible but is heavily influenced by seasonal effects.

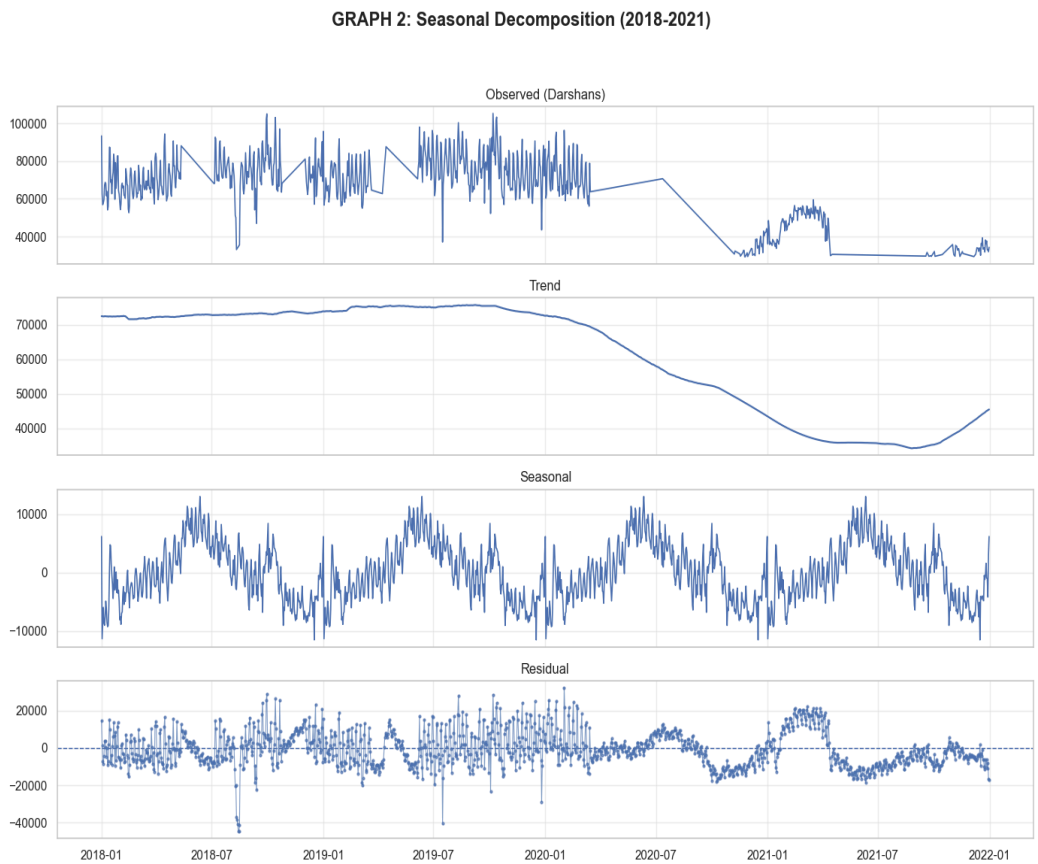
The **Trend** component reveals a consistent long-term growth from around 59,000 visitors per day in early 2014 to approximately 75,000 visitors per day by the end of 2017, indicating a net increase of about 16,000 visitors per day over four years. This upward trend reflects the growing popularity of Tirumala Tirupati Devasthanam, improved infrastructure, and rising religious tourism.

The **Seasonal** component clearly demonstrates strong annual seasonality. Positive peaks of +8k to +12k occur during the summer months (May–June), aligning with the summer flag, school vacations, and major festivals. Negative troughs of –8k to –12k are observed

during the monsoon and winter months (November–December). Sharp spikes within the seasonal component correspond to public holidays, festivals, and Brahmostavam events captured by the binary flags in the dataset.

The **Residual** component shows mostly random fluctuations around zero, with values largely contained within ±25k. The relatively low and random nature of residuals indicates that the trend and seasonal components successfully capture the majority of systematic variation in the data. Occasional larger residuals are attributable to weather-related factors (rainfall) and short-term unmodeled events.

6.2.2 Seasonal Decomposition Analysis (2018–2021)



The **Observed** series shows normal high volatility with daily counts ranging between 60k and 100k+ until early 2020. A drastic collapse occurs from March–April 2020 onward,

with darshans falling sharply to approximately 30k–40k and remaining at this low level throughout 2020–2021.

The **Trend** component reveals a stable level around 68k–70k visitors per day until early 2020. From mid-2020, a sharp downward structural break is evident, with the trend declining steadily to approximately 38k by the end of 2021. This reflects the long-term effect of COVID-19 lockdowns, restricted travel, and temple closures on pilgrim footfall.

The **Seasonal** component continues to exhibit the same strong annual seasonality observed in previous years. Positive peaks of +8k to +12k occur consistently during the summer months (May–June), driven by vacations, the summer flag, and major festivals. Negative troughs of –8k to –12k appear during the monsoon and winter months (November–December). Sharp spikes within the seasonal panel align precisely with the is_festival, is_brahmostavam, and is_public_holiday indicators, confirming that the yearly cyclical pattern persisted even during the pandemic.

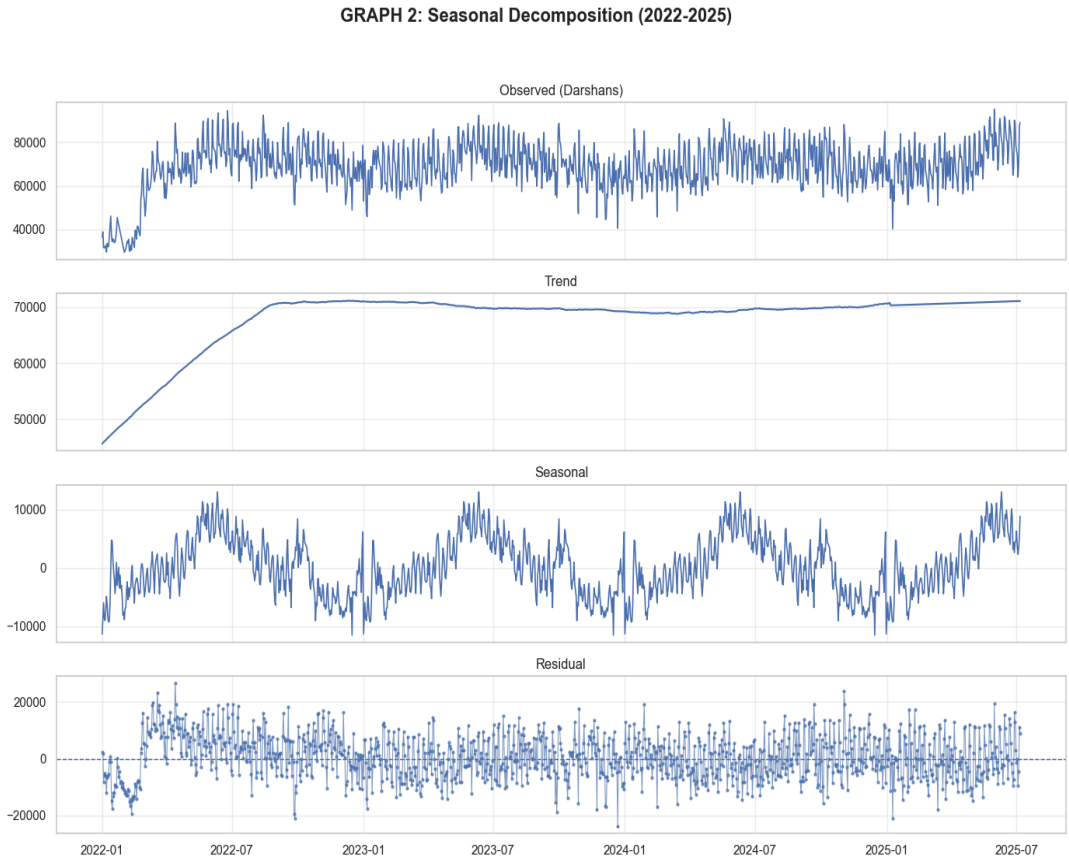
The **Residual** component is the same as above.

6.2.3 Seasonal Decomposition Analysis (2022–2025)

The seasonal decomposition of daily darshans for the post-COVID recovery period 2022–2025 was performed using an additive model with a yearly period of 365 days. This window clearly illustrates the strong rebound in pilgrim footfall after the pandemic.

The **Observed** series begins at a relatively low level (~40,000–50,000 visitors per day) in early 2022. A rapid recovery occurs throughout 2022, with counts quickly rising above 70,000 and stabilizing between 70,000 and 85,000+ from mid-2022 onward. The high daily volatility and sharp spikes typical of weekend and festival days return fully in this period.

The **Trend** component reveals a sharp upward recovery from approximately 45,000 visitors per day in January 2022 to around 70,000 by mid-2022. After this rebound, the trend becomes largely flat and stable at 68,000–70,000 through 2023–2025, with a very slight upward drift visible again in 2025. This reflects the successful reopening of tourism, improved infrastructure, and resumption of long-term growth at Tirumala Tirupati Devasthanam.

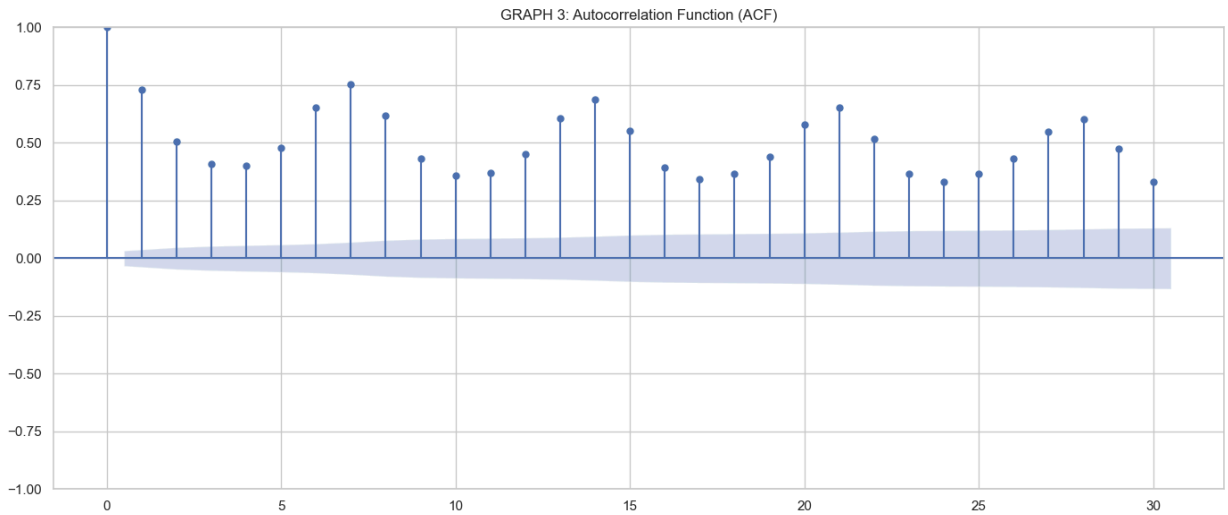


The **Seasonal** component demonstrates that the strong annual seasonality observed in earlier periods is fully restored. Positive peaks of +8,000 to +12,000 consistently occur during the summer months (May–June), driven by school vacations, the summer flag, and major festivals. Negative troughs of –8,000 to –12,000 appear during the monsoon and winter months (November–December). Sharp spikes within the seasonal panel continue to align precisely with the is_festival, is_brahmostavam, and is_public_holiday indicators.

The **Residual** component is the cleanest among all three windows, with most values contained within ±20,000 and highly random around the zero line. This indicates that the trend and seasonal components successfully capture the majority of systematic variation in the post-recovery data, leaving only minor unexplained daily fluctuations.

6.3 Autocorrelation Analysis (ACF and PACF)

1. ACF plot



The ACF plot shows the correlation between the current day's darshans and its past values at different lags. The formula used for the sample autocorrelation at lag k k k is:

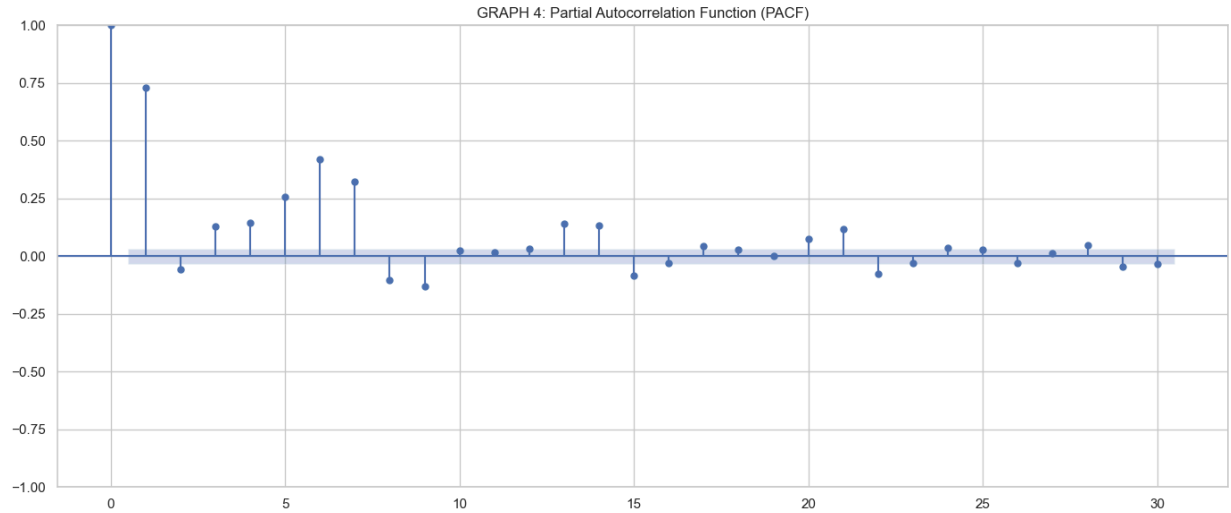
$$r_k = \frac{\sum_{t=1}^{n-k} (y_t - \bar{y})}{\sum (y_t - \bar{y})^2}$$

Key Observations:

- A very high spike at **lag 7, 14, 21, and 28** confirms **strong weekly seasonality**.
- Slow decay of autocorrelation indicates the presence of a **long-term trend** (non-stationary series).

This result is consistent with the weekly pattern observed in the seasonal decomposition.

2. PACF plot



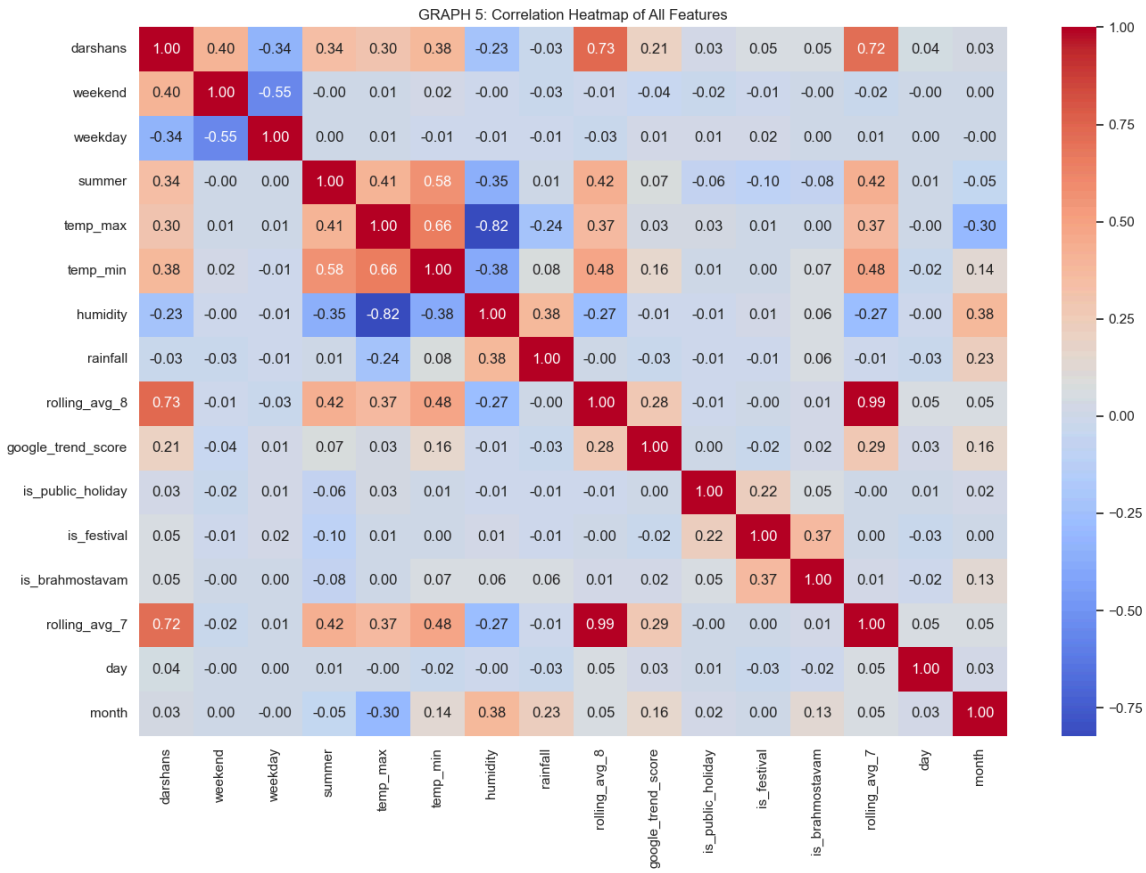
The Partial Autocorrelation Function (PACF) measures the direct relationship between the current day's darshans and its lagged values after removing the effects of all intermediate lags. The formula for the partial autocorrelation coefficient at lag k k k (ϕkk \phi_{kk} ϕkk) is:

$$\phi_{kk} = \frac{r_k - \sum_{j=1}^{r_k} \phi_{kj}\, r_{k-j}}{1 - \sum_{j=1}^{\tilde{k}} \phi_{kj}^{k}\, r_j}$$

Key Observations:

- A very strong spike at **lag 1** (≈ 0.75) indicates that today's darshans are strongly and directly influenced by yesterday's value.
- Small but noticeable spikes appear at **lag 7 and lag 8**, confirming the weekly seasonality effect even after accounting for shorter lags.
- After lag 8, most values fall inside the confidence band and are close to zero, suggesting that higher lags have very little direct influence.

6.4 Correlation Heatmap Analysis



The correlation heatmap presents the Pearson correlation coefficients between the target variable darshans and all explanatory features, revealing the strength and direction of their linear relationships. The strongest positive correlations with daily darshans are observed with the engineered rolling averages — rolling_avg_8 (0.73) and rolling_avg_7 (0.72). This indicates that recent visitor momentum has the greatest influence on today's footfall; in real life, when more pilgrims visit Tirumala over the previous week, it creates a "snowball effect" through word-of-mouth and increased awareness, leading to even higher darshans in the following days.

The weekend flag shows a moderately strong positive correlation (0.40), while the weekday flag exhibits a moderate negative correlation (−0.34). This clearly reflects the real-world behaviour of devotees who prefer to undertake the pilgrimage on weekends when they are free from work and school commitments. The summer season variable is

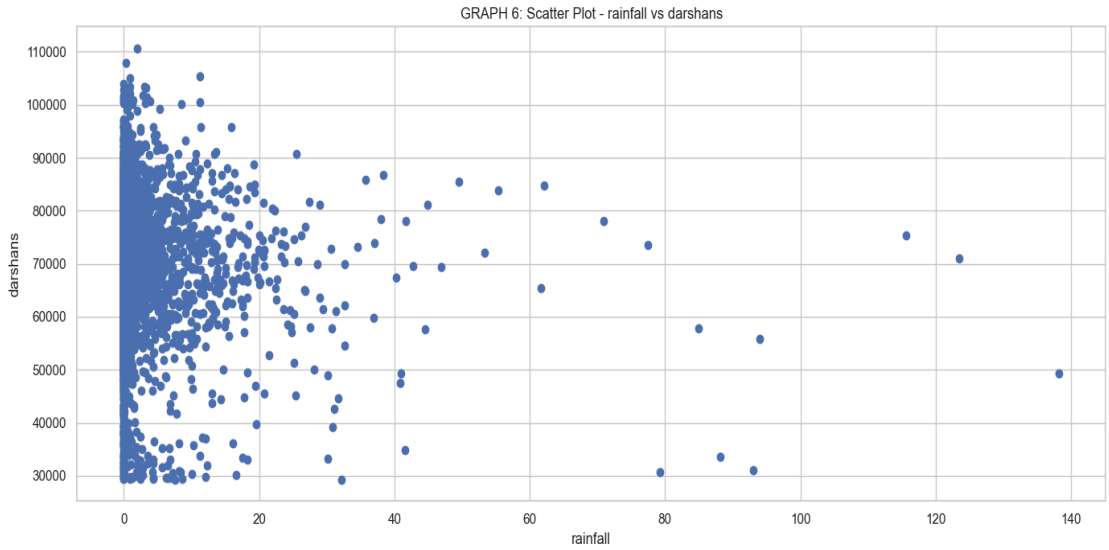
positively correlated (0.34) with darshans, supported by the positive correlation of temp_max (0.30). During summer months, school vacations and favourable weather encourage large family pilgrimages to Tirumala, resulting in significantly higher daily visitor counts.

Among the event-related features, is_brahmostavam demonstrates a relatively strong positive correlation (0.37), followed by is_festival (0.22) and is_public_holiday (0.05). These spikes occur because major religious festivals and holidays draw massive crowds from across India who plan their visit around these auspicious occasions. In contrast, humidity shows a moderate negative correlation (−0.23), implying that high humidity and discomfort during the monsoon season tend to discourage pilgrims. The Google_trend_score has a modest positive correlation (0.21), suggesting that increased online interest and search activity often precedes higher physical footfall at the temple.

Overall, the heatmap confirms that **recent visitor momentum (rolling averages), weekend effect, summer season, and major festivals** are the most influential predictors of daily darshans at Tirumala Tirupati Devasthanam. These findings align well with practical observations of pilgrimage behaviour and provide strong guidance for building accurate forecasting models.

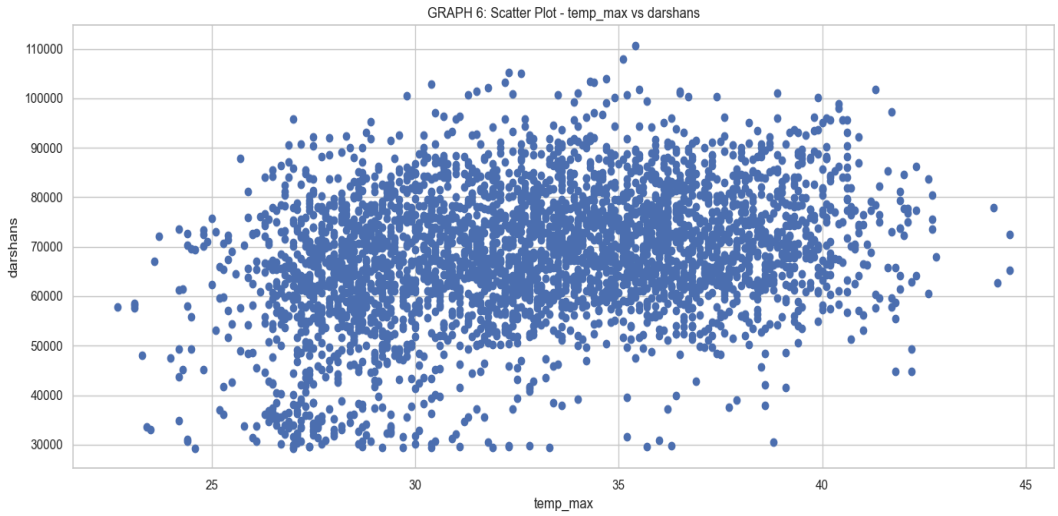
6.5 Scatter Plot Analysis

6.5.1 Rolling Averages Effect (rolling_avg_7 and rolling_avg_8)

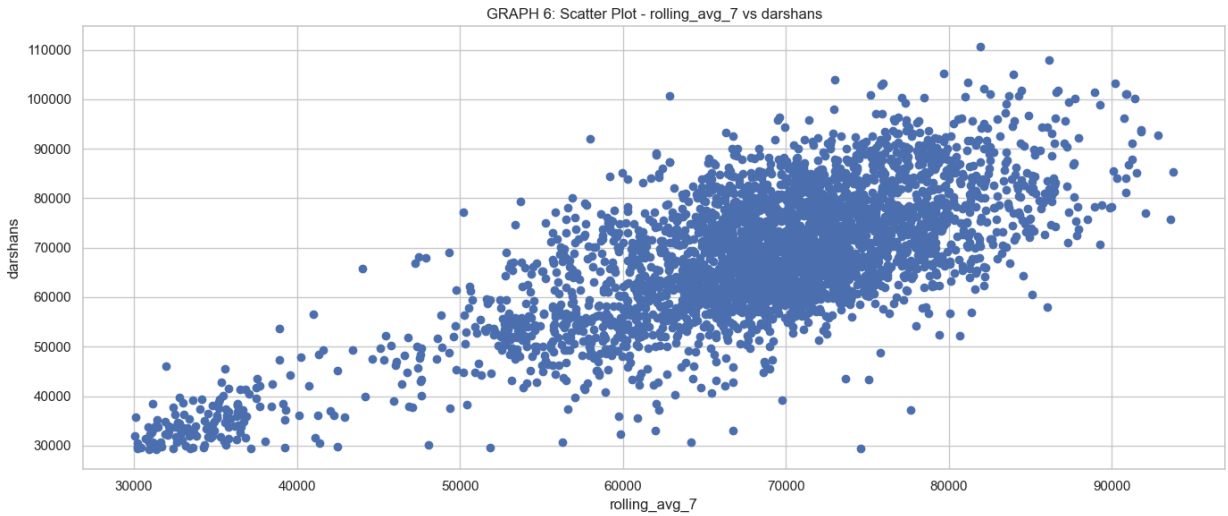


The feature rainfall shows a weak negative correlation of -0.03 with the target variable darshans. This indicates that as rainfall increases, the number of daily visitors tends to decrease. In real-world terms, this relationship confirms the decline in darshans during the monsoon season, when heavy rainfall and unfavourable weather conditions discourage pilgrims from undertaking the journey to Tirumala.

6.5.2 Temperature Effect (temp_max and temp_min)



6.5.3 Rainfall Effect



The scatter plot (Graph 6) visualizes the relationship between the 7-day rolling average of darshans (rolling_avg_7) and the actual daily number of darshans.

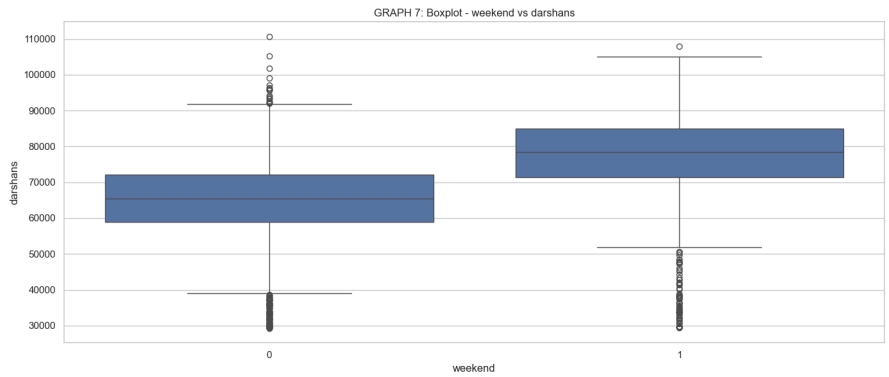
A **strong positive linear relationship** is clearly visible. As the 7-day rolling average increases from approximately 30,000 to 95,000, the daily darshans also rise consistently from around 30,000 to over 110,000. The points are tightly clustered along an upward-sloping trend, confirming the high positive correlation (0.72) observed in the correlation heatmap.

Interpretation and Real-Life Insight

This strong positive relationship indicates that **recent visitor momentum has a powerful influence** on current-day footfall. In practical terms, when more pilgrims visit Tirumala over the previous seven days, it creates a “snowball effect” through word-of-mouth, increased social media visibility, and heightened awareness among devotees. This momentum encourages even more people to plan their visit in the following days.

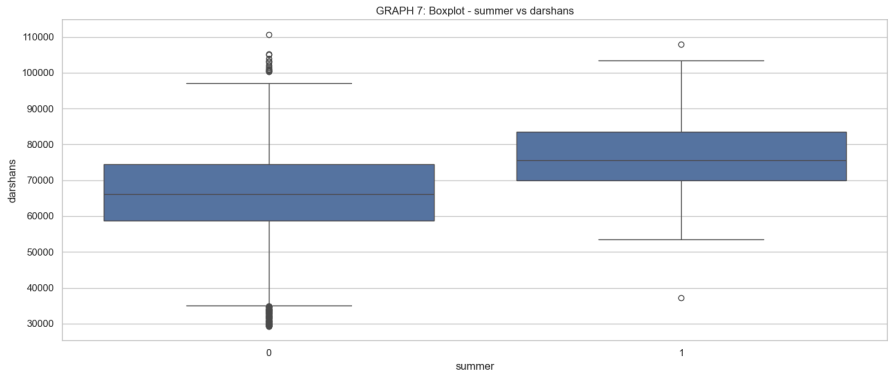
6.6 Boxplot Analysis

6.6.1 Weekend Effect



The boxplot (Graph 7) shows the distribution of daily darshans on weekdays versus weekends. When weekend = 1, the median darshans is substantially higher and the entire box shifts upward compared to weekdays. This confirms a strong weekend effect, where visitor numbers increase noticeably on Saturdays and Sundays. In practice, devotees tend to plan their pilgrimage on weekends when they have more free time, leading to higher darshans at Tirumala Tirupati Devasthanam.

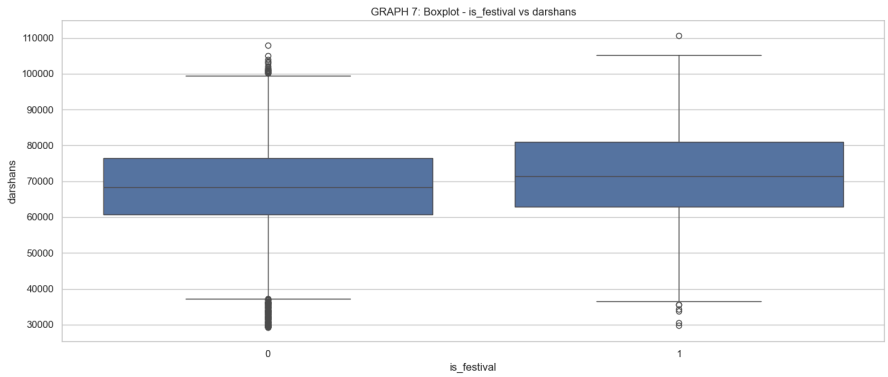
6.6.2 Summer Season Effect



The boxplot shows the distribution of daily darshans during summer season (is_summer = 1) versus non-summer months (is_summer = 0).

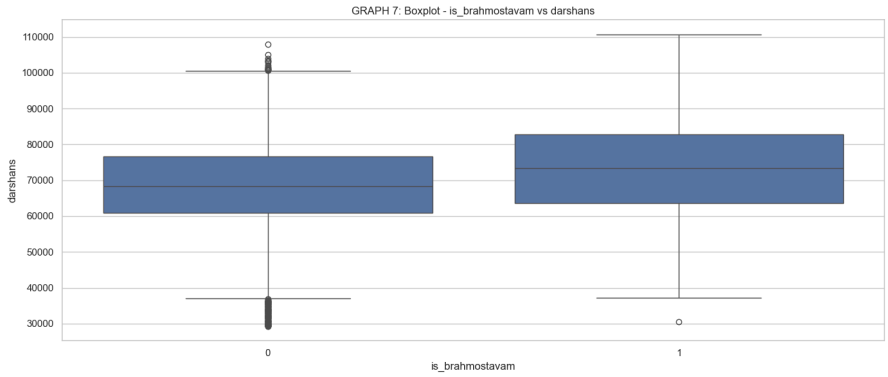
During summer months, the median darshans is noticeably higher, and the entire box shifts upward compared to non-summer periods. This confirms that the **summer season has a strong positive impact** on daily visitor footfall. In reality, summer coincides with school vacations, longer holidays, and favourable travel weather, encouraging large numbers of families and groups to visit Tirumala.

6.6.3 Festival Effect (is_festival)



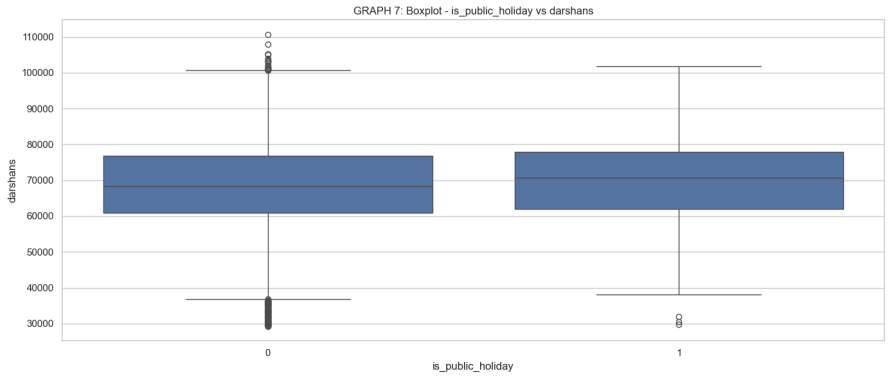
The boxplot (Graph 7) illustrates the distribution of daily darshans on normal days versus festival days. When is_festival = 1, the median number of darshans is noticeably higher than on normal days. The box shifts upward, showing an increase in visitor numbers during festival periods. This confirms that festivals have a positive, though moderate, effect on darshans at Tirumala Tirupati Devasthanam, as many devotees choose these auspicious days to undertake the pilgrimage.

6.6.4 Brahmostavam Effect (is_brahmostavam)



The boxplot (Graph 7) shows the distribution of daily darshans on normal days versus Brahmostavam festival days. When is_brahmostavam = 1, the median number of darshans rises noticeably higher than on normal days (0). The entire box shifts upward, indicating a significant increase in visitor footfall during the Brahmostavam period. This confirms that Brahmostavam is one of the strongest event-based drivers of darshans at Tirumala Tirupati Devasthanam, as devotees flock to the temple in large numbers during this highly auspicious festival.

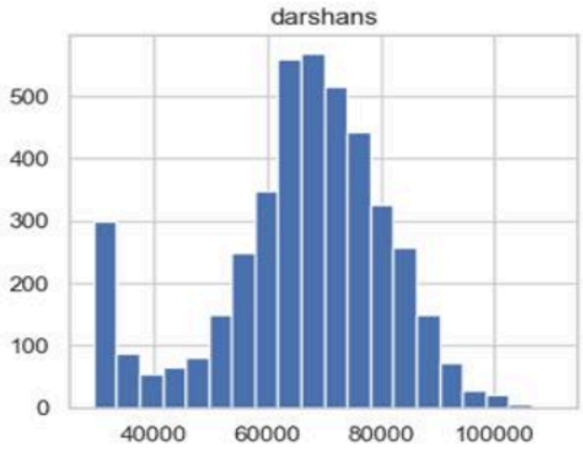
6.6.5 Public Holiday Effect (is_public_holiday)



The boxplot compares the distribution of daily darshans on normal days versus public holiday days. On public holidays (is_public_holiday = 1), the median darshans is slightly higher compared to normal days. The overall distribution shows a modest upward shift, indicating that public holidays have a **mild positive effect** on visitor numbers. However, the increase is not as strong as during major festivals or Brahmostavam. In real life, while holidays provide more free time for travel, many devotees prefer to visit on weekends or auspicious festival days rather than general public holidays.

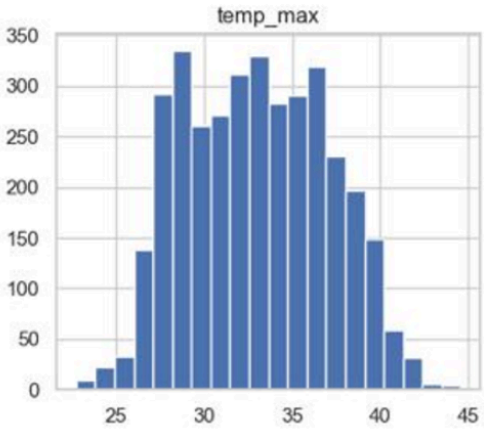
6.7 Histogram Distribution Analysis

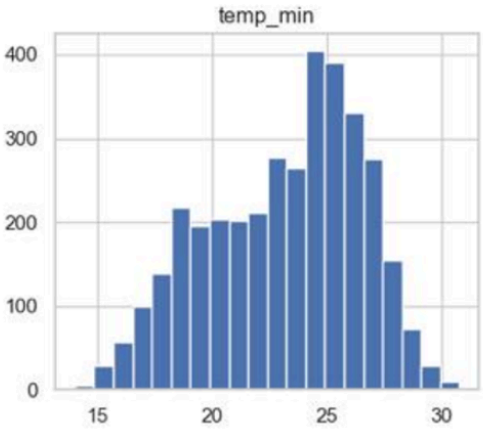
6.7.1 Distribution of Daily Darshans



The histogram of daily darshans shows a unimodal, slightly right-skewed distribution. Most values are concentrated between 60,000 and 85,000 visitors per day, with the highest frequency around 70,000–75,000. The distribution has a long tail on the right side, indicating occasional high peaks (above 95,000–110,000) on special days such as festivals, Brahmostavam, or peak weekends. Very few days have extremely low darshans (below 40,000), except during the COVID-19 period.

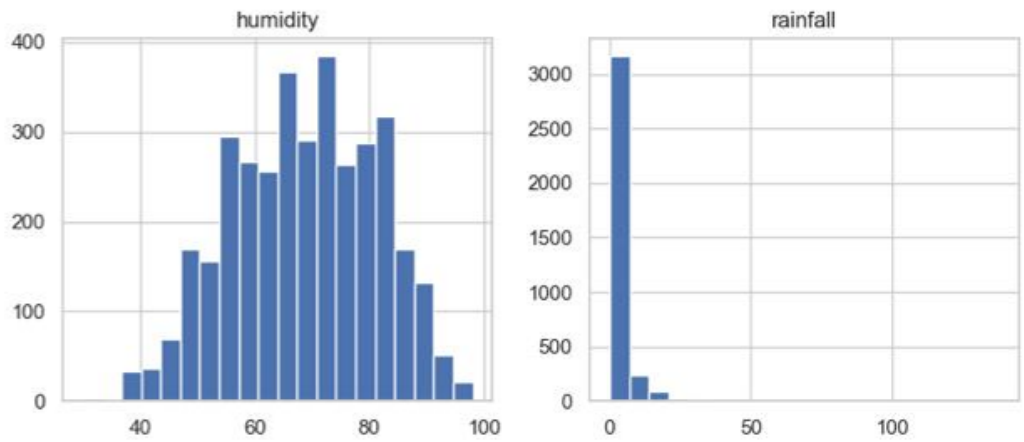
6.7.2 Distribution of Temperature (temp_max and temp_min)





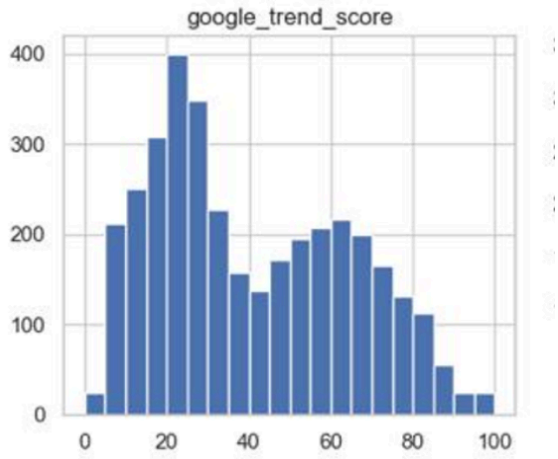
The histograms of maximum and minimum temperatures reveal that the weather in the Tirumala region is generally warm throughout the year. The maximum temperature (temp_max) mostly ranges between 28°C and 38°C, while the minimum temperature (temp_min) typically lies between 20°C and 28°C. Both distributions are relatively normal with mild skewness, indicating consistent climatic conditions without extreme variations. This moderate temperature range supports higher pilgrim footfall, especially during the summer season when people have more free time for travel.

6.7.3 Distribution of Humidity and Rainfall



The histograms of humidity and rainfall show contrasting patterns. Humidity levels are normally distributed, with most days experiencing moderate to high humidity between 60% and 85%. In contrast, the rainfall distribution is heavily right-skewed, with the majority of days recording zero or negligible rainfall and only a few days with significant rain. This confirms that while the region maintains consistently high humidity, actual rainfall occurs infrequently and is mostly concentrated during the monsoon months, which may negatively affect daily darshans on rainy days.

6.7.4 Distribution of Google Trends Score



The histogram of google_trend_score exhibits a bimodal distribution with peaks around 20–30 and 60–80. This indicates that search interest for Tirumala is mostly moderate, with occasional significant surges. These surges likely correspond to major festivals, Brahmostavam, or public holidays, which drive higher online awareness and subsequently influence physical visitor turnout. The distribution supports the moderate positive correlation (0.21) between Google Trends and daily darshans.

7. Analysis Summary:-

The Exploratory Data Analysis provided valuable insights into the relationships between various features and the target variable, Darshans. Several key observations emerged from the visualizations and statistical analysis:

- Scatter plots were found to be less effective for categorical and binary variables (such as weekend, summer, is_festival, is_brahmostavam, and is_public_holiday). Therefore, boxplots were used instead, as they offer a clearer comparison of the distribution of darshans across different categories and help identify shifts in median values and variability.
- Histogram and correlation analysis revealed that some features, particularly is_public_holiday and is_brahmostavam, exhibit very weak correlation with daily darshans. While is_brahmostavam showed a moderate correlation, its impact is limited to only a few days annually and is largely captured by the broader is_festival flag and seasonal patterns.
- These low-correlation features may introduce unnecessary noise into the model, potentially reducing its efficiency and predictive performance without adding meaningful information.

Decision on Feature Selection

Based on the above analysis, the following low-importance or redundant features were removed from the final modeling dataset:

- rolling_avg_7 (due to high multicollinearity with rolling_avg_8)
- weekday (redundant with weekend)
- is_public_holiday
- is_brahmostavam

These features will not be included in the final forecasting model unless subsequent testing (such as XGBoost or Random Forest feature importance) demonstrates significant predictive value. This selective approach ensures a cleaner, more focused, and more efficient model while retaining the most influential variables, such as rolling averages, the weekend effect, the summer season, and festival indicators.

8. Conclusion :-

The Tirumala Tirupati Devasthanams daily darshan dataset is really interesting. It gives us a lot of information about the patterns and key factors that influence the number of pilgrims visiting the Tirumala Venkateswara Temple. The Tirumala Venkateswara Temple is one of the religious places in the world.

If we look at the data, we can see that there are more pilgrims on weekends than on weekdays. This is a weekly pattern. We also see that during the summer, there are pilgrims. This is probably because schools are on vacation, and it is easier to travel. The 8-day rolling average of pilgrim numbers shows us that the number of pilgrims today is related to the number of pilgrims over the past few days. This means that if many pilgrims visit the temple one day, it is likely that many will visit the temple the next day too. Big festivals like Brahmotsavam also bring a lot of pilgrims to the temple.

The weather also affects the number of pilgrims. When it is hot, more people come to the temple.. When it rains, fewer people come. We also looked at Google Trends data, which shows what people are searching for online. We found that people often search for information about the temple before they visit. The relationship between online searches and actual visits is not very strong.

When we broke down the data by season, we saw that the pattern of pilgrim numbers is similar every year. There are always pilgrims in the summer and during festivals.. During the COVID-19 period from 2020 to 2021, the pattern was different. This is because many people could not travel due to the pandemic.

We removed some variables from the data that were not very useful. These included the 7-day average, whether it was a weekday or not, whether it was a public holiday, and whether it was Brahmotsavam or not. By removing these variables, we made the data more robust and easier to understand.

Overall, our analysis of the Tirumala Tirupati Devasthanams daily darshan dataset has given us a lot of information about what affects the number of pilgrims visiting the Tirumala Venkateswara Temple. This information can be used to make decisions about how to manage the crowds, allocate resources, and plan operations at the temple. The Tirumala Tirupati Devasthanams daily darshan dataset is a tool for the temple administration.